

A Visual Reference for Mathematical and Scientific LaTeX

Patterns for equations, tables, plots, diagrams, layouts, and research communication

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Abstract

This document is a self-contained visual catalog of LaTeX patterns for research writing. It shows how mathematical notation, aligned derivations, theorem-like statements, tables, plots, diagrams, algorithms, code, annotations, and page layouts can be expressed in a polished PDF. Each section is organized by communication goal: choose a pattern by asking what the reader needs to see, compare, trace, compute, or remember. The examples use standard LaTeX packages, especially `amsmath`, `booktabs`, `TikZ`, `PGFPlots`, and `listings`.

Skim the headings and captions as a menu of visual strategies. When a design resembles what you need, reuse the associated package family: equations with `amsmath`, tables with `booktabs`/`tabularx`, plots with `PGFPlots`, diagrams with `TikZ`, and code with `listings`. The examples are intentionally compact, so each page functions like a reference sheet rather than a tutorial.

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1 Visual Design Principles for Research PDFs

LaTeX is strongest when content and structure are encoded semantically. A useful reference PDF should therefore separate *what is being communicated* from *how it is styled*. The same mathematical result can be shown as an equation, a table, a flow diagram, a plot, or a boxed rule depending on the reader's task.

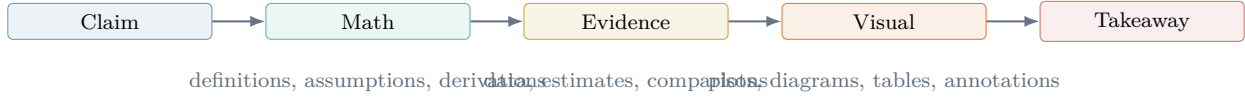


Figure 1: A research-paper visual sequence: make the claim explicit, define the mathematics, present evidence, then encode the takeaway visually.

- **Explain** Use aligned equations, diagrams, and call-outs.
- **Quantify** Use axes, uncertainty bands, error bars, and units.
- **Compare** Use tables, grouped plots, heatmaps, and small multiples.
- **Remember** Use boxed rules, summaries, and consistent color coding.
- **Trace** Use flowcharts, timelines, trees, and causal graphs.
- **Navigate** Use sectioning, captions, labels, and a pattern index.

2 Mathematical Communication

Mathematics in LaTeX is more than typeset symbols: it is spatial organization. Use alignment, grouping, numbering, color, and annotations to reveal logical structure.

2.1 Equation Forms

The alignment point & acts like a visual spine. It lets the eye compare equal signs, assumptions, transformations, and justifications.

$$\mathcal{L}(\theta) = \sum_{i=1}^n \log p_{\theta}(y_i | x_i) - \lambda \|\theta\|_2^2 \quad \text{regularized objective} \quad (1)$$

$$\nabla_{\theta} \mathcal{L}(\theta) = \sum_{i=1}^n \nabla_{\theta} \log p_{\theta}(y_i | x_i) - 2\lambda \theta \quad \text{first-order condition} \quad (2)$$

$$\hat{\theta} = \arg \max_{\theta \in \mathbb{R}^d} \mathcal{L}(\theta). \quad \text{estimator} \quad (3)$$

$$f(x) = \begin{cases} 0, & x < 0, \\ x, & 0 \leq x \leq 1, \\ 1, & x > 1, \end{cases} \quad \hat{x} = \arg \min_{x \in \mathbb{R}^n} \frac{1}{2} \|Ax - b\|_2^2 + \lambda \|x\|_1. \quad (4)$$

$$\boxed{\mathbb{E}[Y | X = x] = \int_{-\infty}^{\infty} y p(y | x) dy} \quad \overbrace{\frac{1}{n} \sum_{i=1}^n \ell(y_i, \hat{y}_i)}^{\text{empirical loss}} + \underbrace{\lambda \Omega(\theta)}_{\text{complexity penalty}}. \quad (5)$$

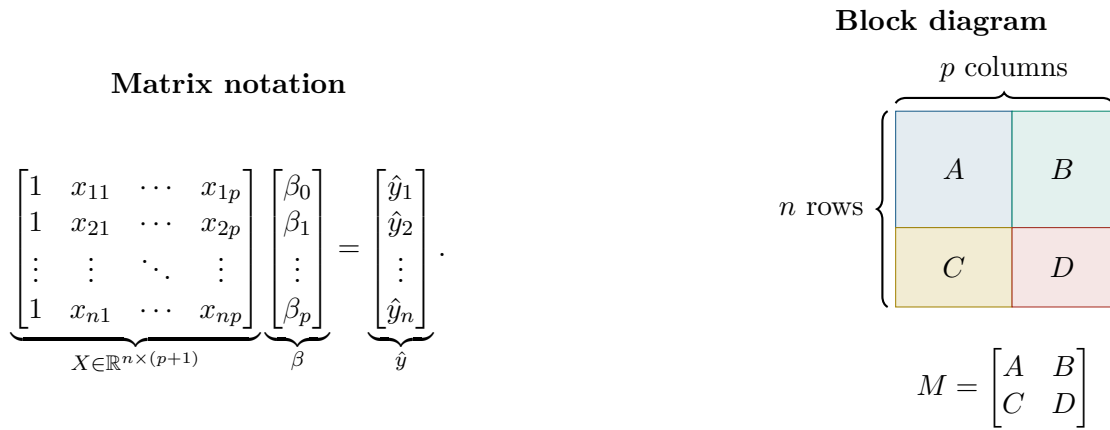


Figure 2: Matrices can be shown symbolically or as spatial block diagrams. The diagram is often better when dimensions, partitions, or sparse structure matter.

2.2 Matrices, Vectors, and Block Structure

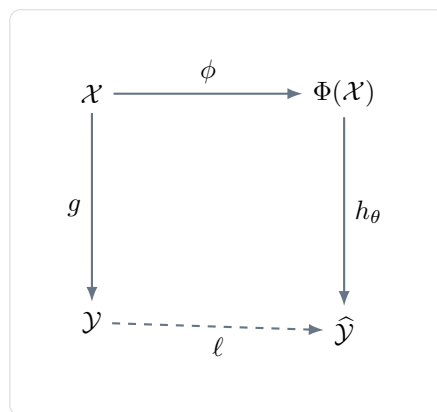
2.3 Definitions, Theorems, and Proofs

Definition 2.1 (Convex set). A set $C \subseteq \mathbb{R}^d$ is convex if $tx + (1 - t)y \in C$ for all $x, y \in C$ and all $t \in [0, 1]$.

Theorem 2.2 (Projection uniqueness). *If $C \subseteq \mathbb{R}^d$ is nonempty, closed, and convex, then every point $z \in \mathbb{R}^d$ has a unique Euclidean projection onto C .*

Proof. Existence follows from continuity of $x \mapsto \|z - x\|_2$ on a closed bounded level set. If two distinct minimizers existed, strict convexity of the squared norm along the line segment joining them would produce a closer feasible point, contradicting minimality. $\square$

2.4 Commutative and Relational Diagrams



Use diagrams like this when maps and compositions matter.

Figure 3: A TikZ commutative-style diagram emphasizes transformations, spaces, and loss functions.

3 Tables and Structured Comparisons

Tables are visual arguments. Use them when exact values, categorical comparisons, or compact summaries matter more than trend shape.

Model	Accuracy	F1	Parameters	Time	Best use
Linear model	0.81	0.78	1.2×10^3	0.2 s	transparent baseline
Random forest	0.88	0.85	4.8×10^5	3.7 s	mixed tabular data
Neural network	0.92	0.90	2.1×10^6	11.4 s	high-capacity model
Gaussian process	0.86	0.84	n^2 kernel	9.8 s	calibrated uncertainty

Table 1: A clean comparison table uses few rules, clear columns, consistent precision, and a short interpretation column.

3.1 Booktabs Tables

3.2 Heatmaps, Sparklines, and Encoded Cells

System	Robustness	Speed	Memory	Trend	Verdict
Alpha	0.72	0.41	0.80		steady
Beta	0.84	0.76	0.48		best
Gamma	0.55	0.91	0.63		fast

Table 2: Cells can carry more than text: background color, badges, and sparklines make a table glanceable without hiding the numbers.

3.3 Reference Tables with Flexible Columns

Pattern	Use when	Primary LaTeX tools
Aligned derivation	The reader must audit each algebraic step.	<code>align</code> , <code>aligned</code> , <code>split</code> from <code>amsmath</code> .
Piecewise rule	A definition changes across regimes, cases, or thresholds.	<code>cases</code> , relational symbols, explanatory text.
Comparison table	Exact values and rankings matter.	<code>booktabs</code> , <code>tabularx</code> , <code>xcolor</code> .
Process diagram	Sequence, branching, or dependency matters.	<code>TikZ</code> nodes, arrows, positioning.
Quantitative plot	Shape, scale, uncertainty, or trend matters.	<code>PGFPlots</code> axes, marks, legends, grids.

4 Quantitative Plots

Plots convert numeric comparisons into spatial comparisons. In LaTeX, `PGFPlots` produces vector graphics with fonts and colors that match the surrounding document.

5 Diagrams for Models, Processes, and Systems

`TikZ` is the general-purpose drawing engine for vector diagrams. It is useful when the visual object is conceptual rather than purely data-driven.

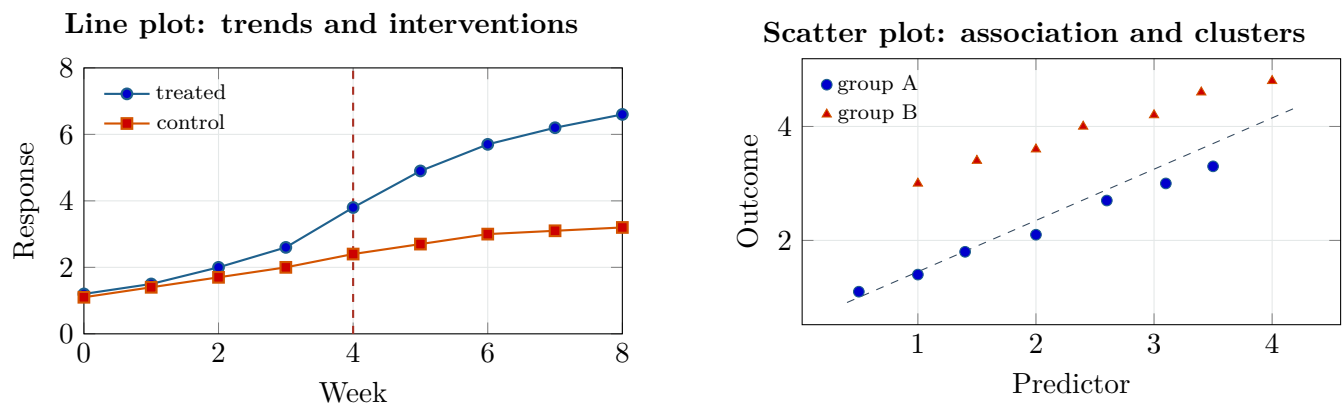


Figure 4: Use line plots for temporal trajectories and scatter plots for relationships, outliers, or clusters.

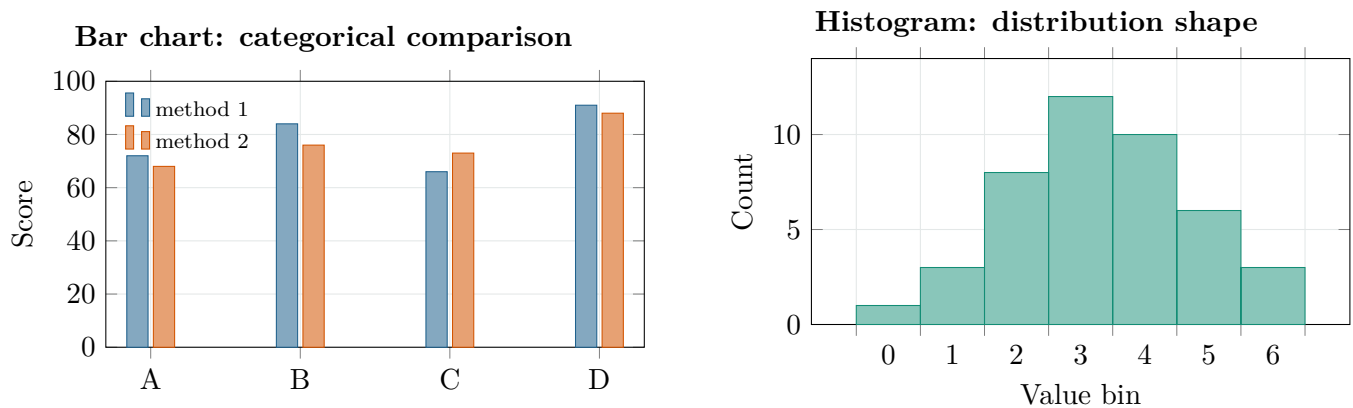


Figure 5: Use bars for categories and histograms for distributions. Keep axis labels explicit so the visual encoding is unambiguous.

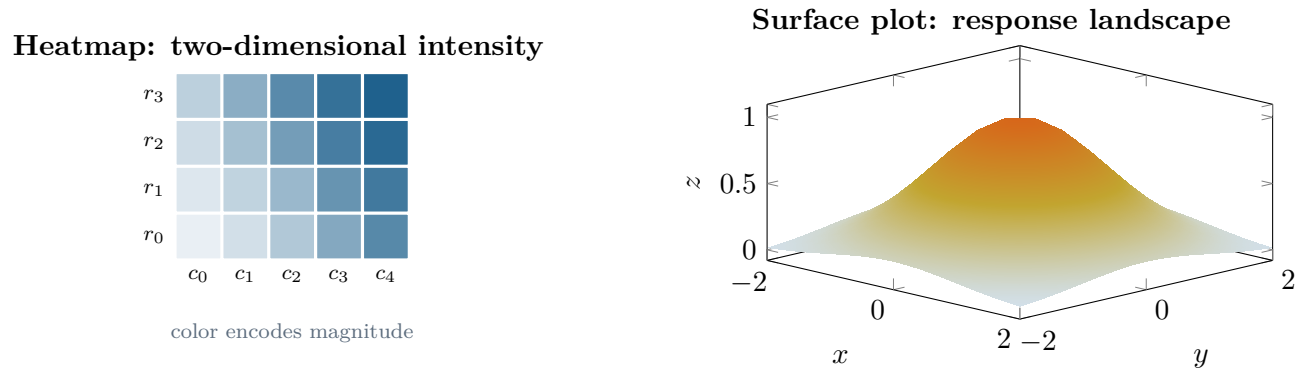


Figure 6: Use heatmaps for matrix-like quantities and surfaces for functions over two continuous variables.

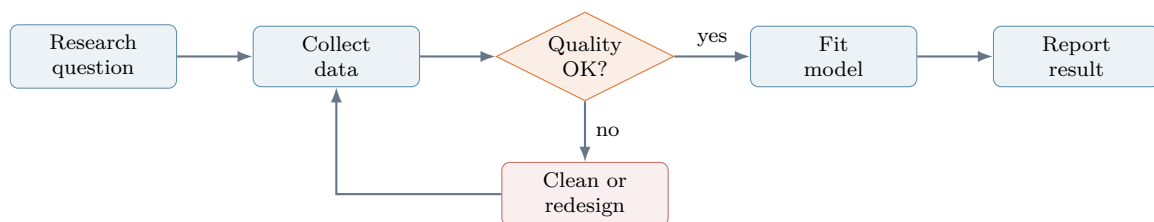


Figure 7: Flowcharts show sequence and branching. Use a diamond for decisions and keep labels action-oriented.

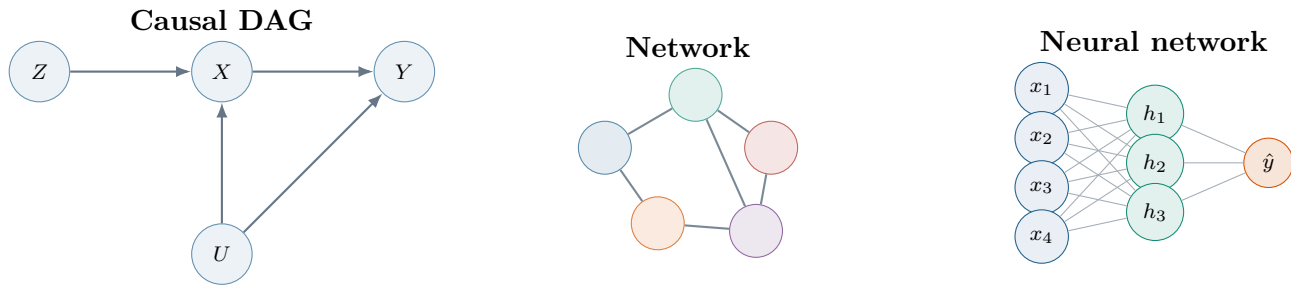


Figure 8: Conceptual diagrams expose dependencies, connectivity, and architecture more directly than prose.

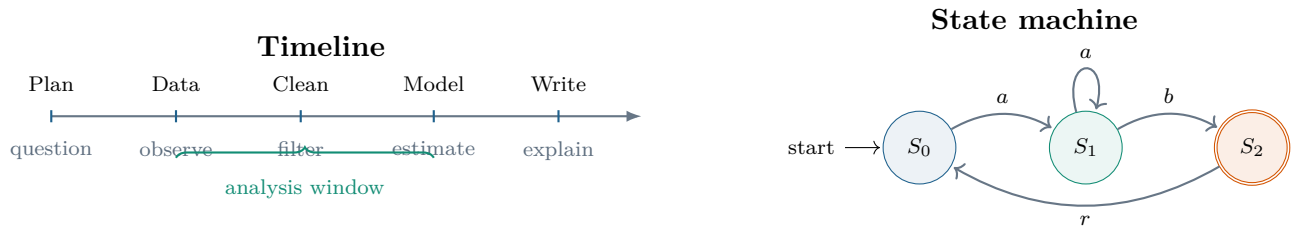


Figure 9: Timelines emphasize ordering and duration; state machines emphasize transitions and recurrence.

5.1 Flowcharts and Pipelines

5.2 Causal Graphs, Networks, and Architectures

5.3 Timelines, Trees, and State Machines

6 Geometric, Spatial, and Annotated Visuals

Spatial diagrams are useful when the object of study has geometry: regions, distances, transformations, angles, fields, images, or physical layout.

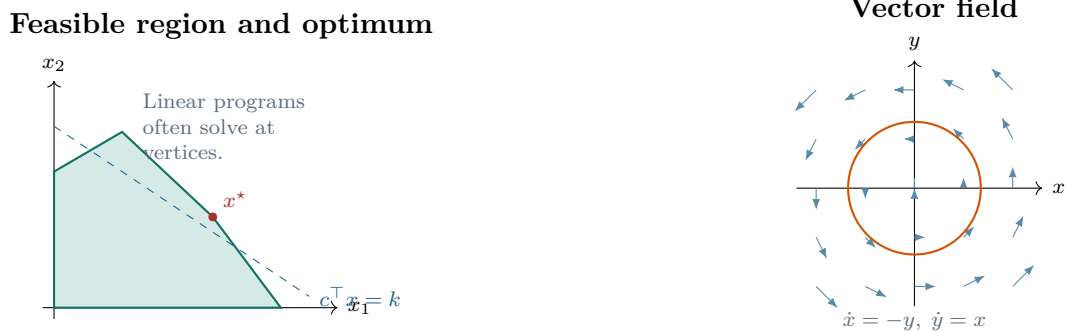


Figure 10: Coordinate drawings can combine axes, shaded regions, constraints, annotations, and vector arrows.

7 Algorithms, Code, and Data Workflows

Algorithms are easier to audit when the document separates input, iteration, decision points, and output. Source code is best shown when exact syntax matters; pseudocode is best when the idea matters.

8 Color, Emphasis, and Layout Patterns

Color should encode meaning, not decoration. A compact palette with repeated roles helps readers learn the visual language of the paper.



Figure 11: Visual annotations turn a drawing into an explanation: label sets, dimensions, parts, or mechanisms directly in the figure.

Pseudocode table	Executable-style code listing
1 Initialize parameters θ_0 and learning rate η. 2 For $t = 0, \dots, T-1$, compute $g_t = \nabla_{\theta} \mathcal{L}(\theta_t)$. 3 Update $\theta_{t+1} = \theta_t - \eta g_t$. 4 Stop when $\ g_t\ _2 < \varepsilon$ or the validation score declines.	<pre> 1 def gradient_step(theta, grad, eta): 2 """One step of first-order 3 optimization.""" 4 return theta - eta * grad(theta) </pre>

Figure 12: Use pseudocode for algorithmic logic and listings for exact implementation details.

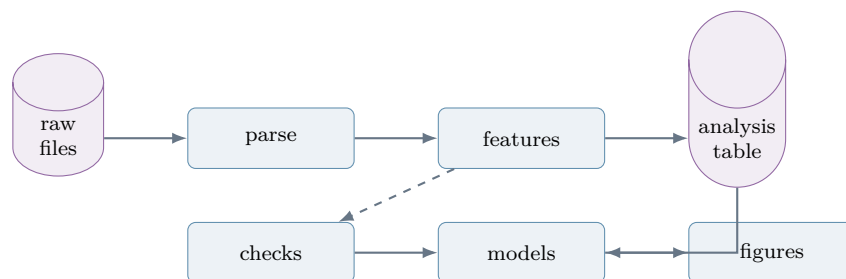


Figure 13: Workflow diagrams are ideal for reproducible research pipelines: data enters, transformations are explicit, checks are visible, and outputs are named.

Role	Swatch	Suggested use
Primary structure		main variables, axes, first condition, main method
Secondary structure		comparison method, positive status, feasible region
Warning or intervention		thresholds, changes, alerts, alternative path
Important exception		errors, rejected cases, critical point, conflict
Context or neutral frame		axes, outlines, background labels, secondary text

Table 3: A role-based color palette is easier to interpret than one-off decorative colors.

Use a boxed card for a definition, rule of thumb, warning, or result that the reader should find quickly. Cards work best when they are short, named, and visually consistent.	Use warm colors sparingly for assumptions, limitations, failed checks, or practical cautions. Too many warnings make every warning look unimportant.
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Figure 14: Boxed text creates navigational landmarks in dense technical documents.

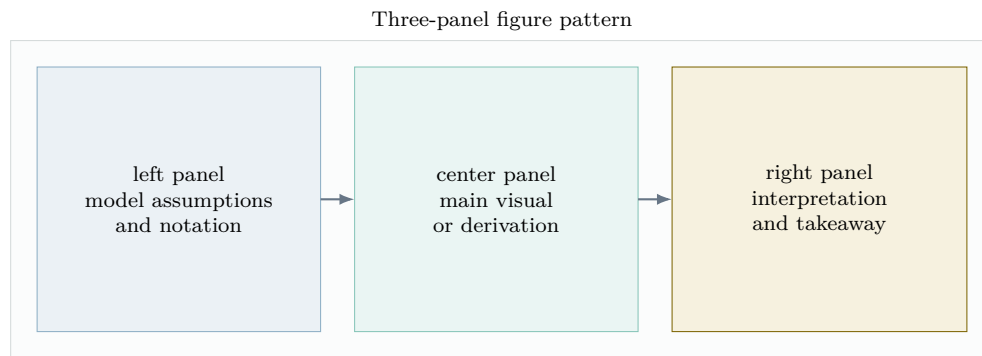


Figure 15: Multi-panel layouts help readers compare input, transformation, and interpretation without turning pages.

9 Package Map and Pattern Index

The table below is a practical index of visual forms. It is not a list of every LaTeX package; it is a decision guide for choosing the right representation.

Visual goal	Good representation	Common tools
Show algebraic logic	Numbered equations, aligned derivations, boxed final result	<code>amsmath</code> , <code>mathtools</code> , <code>bm</code>
Define regimes	Piecewise functions, cases, threshold tables	<code>cases</code> , <code>array</code> , explanatory text
State reusable facts	Definitions, theorems, proofs, examples	<code>amsthm</code> , custom boxed environments
Compare exact values	Tables with grouped rows, units, precision, highlights	<code>booktabs</code> , <code>tabularx</code> , <code>xcolor</code>
Compare trends	Line plots, small multiples, sparklines	<code>PGFPlots</code> , <code>TikZ</code>
Show distributions	Histograms, density plots, box-like summaries	<code>PGFPlots</code>
Show relationships	Scatter plots, regression lines, networks	<code>PGFPlots</code> , <code>TikZ</code>
Show matrices	Heatmaps, block matrices, annotated arrays	<code>TikZ</code> , <code>xcolor</code> , <code>amsmath</code>
Show process	Flowchart, pipeline, state machine, timeline	<code>TikZ</code> with arrows and positioning
Show dependency	Causal DAG, hierarchy, architecture diagram	<code>TikZ</code> nodes, edges, fit boxes
Show geometry	Coordinate diagrams, regions, vector fields, measurements	<code>TikZ</code> , <code>PGFPlots</code>
Show implementation	Code listings, pseudocode tables, workflow diagrams	<code>listings</code> , <code>tabularx</code> , <code>TikZ</code>
Show emphasis	Callouts, badges, summary cards, colored rules	<code>xcolor</code> , custom macros, <code>TikZ</code>

10 Conclusion

LaTeX can function as a complete visual language for research communication. Its core strength is consistency: the same source file can combine precise mathematics, publication-quality tables, vector plots, conceptual diagrams, algorithms, code, and structured page layouts. A strong scientific PDF chooses the simplest visual form that answers the reader's immediate question, then reinforces that choice with clear captions, labels, and repeated visual conventions.